How-To Library

Informatica

Enabling SAML Authentication with Oracle Access Manager for Web Applications

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Abstract

You can enable users to log into Informatica web applications using single sign-on. This article explains how to configure single sign-on in an Informatica 10.5 domain using Security Assertion Markup Language (SAML) v2.0 and the Oracle Access Manager version 12.2.1 identity provider.

Supported Versions

- Informatica Data Engineering Integration 10.5
- Informatica Enterprise Data Catalog 10.5
- Informatica Enterprise Data Preparation 10.5
- Informatica Metadata Manager 10.5

Table of Contents

Overview . . . 3
SAML Authentication Process for Oracle Access Manager. . . . 3
Before You Begin . . . 4
Prerequisites . . . 4
Verify the Oracle Access Manager Environment . . . 4
Configure Oracle Access Manager. . . . 5
Step 1. Configure Active Directory as the Identity Store . . . 5
Step 2. Install and configure Oracle HTTP Server and Web Gate. . . . 5
Step 3. Create a Service Provider Attribute Profile. . . . 5
Step 4. Create an Authentication Module. . . . 6
Step 5. Create an Authentication Scheme. . . . 6
Step 6. Create a Service Provider Connection. . . . 7
Step 7. Configure an Authentication Method and Certificate . . . 7
Step 8. Add the Informatica Domain to the Hosts File. . . . 8
Enable Oracle Access Manager for the Informatica Domain. . . . 9
Edit the hosts/etc File on Domain and Local Machines. . . . 9
Import the Truststore Certificate. . . . 9
Update the Domain SAML Configuration. . . . 9
Optional: Configure the Certificate Alias on Gateway Nodes. . . . 9
Sync Oracle Access Manager LDAP Users. . . . 9
Complete Domain Configuration. . . . 10

Overview

You can configure Security Assertion Markup Language (SAML) authentication in an Informatica 10.5 domain using the Oracle Access Manager identity provider.

An identity provider is an entity that provides authentication as a consumable service by applications. Platforms like Amazon Web Services (AWS) and Microsoft Azure support various third-party identity providers to authenticate requests by applications on their platforms.

SAML is an XML-based data format for exchanging authentication information between a service provider and an identity provider. In an Informatica domain, an Informatica web application is the service provider.

You can configure the following Informatica web applications to use SAML authentication:

- Informatica Administrator
- Informatica Analyst
- Metadata Manager
- Enterprise Data Catalog
- Enterprise Data Preparation

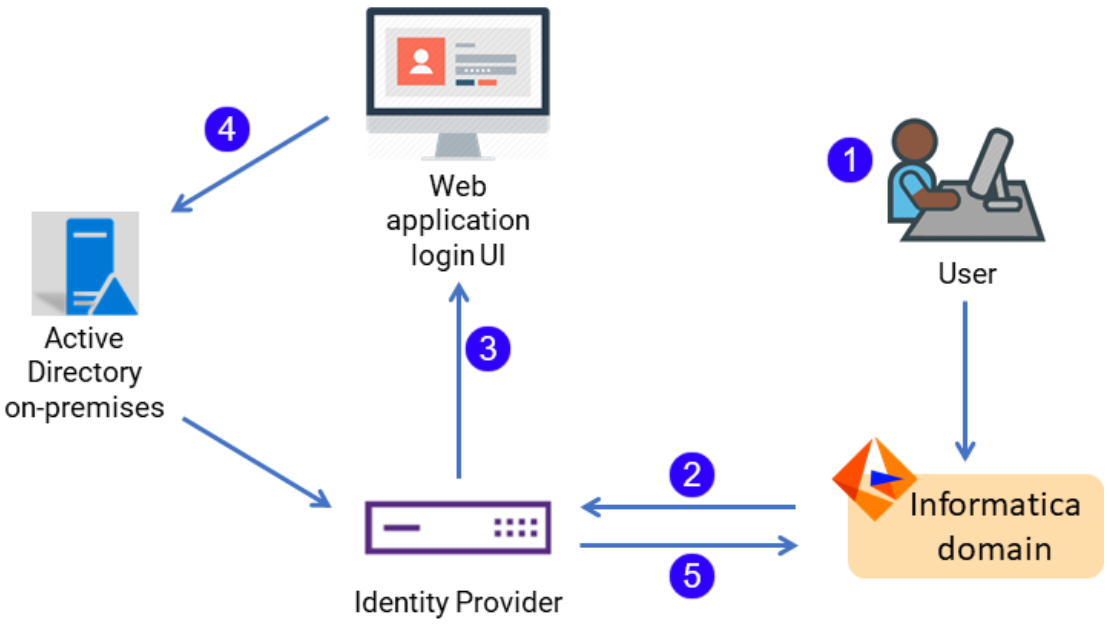
Informatica 10.5 supports Oracle Access Manager 12.2.1.

Note: SAML authentication cannot be used in an Informatica domain configured to use Kerberos authentication.

SAML Authentication Process for Oracle Access Manager

Informatica web applications and Oracle Access Manager identity provider exchange authentication information to enable SAML authentication in an Informatica domain.

The following image illustrates the process flow for SAML authentication for Informatica users:



The following steps correspond to the numbered elements in the illustration:

1. The user uses a URL to access a resource of a service provider. In Informatica, the service provider is the domain, and the resource is a web application that the domain serves. For example, the Administrator tool web UI.
2. The service provider forwards the unauthenticated user to the identity provider (idP) via saml:authnRequest.
3. The identity provider points to its SingleSignOnService URL, so the user must sign in.
4. The IdP checks the entered credentials against a user database such as Active Directory and forms a saml:response about the status of the verification request.
5. The IdP sends the saml:response in the form of an XHTML form back to the service provider. This XHTML form contains, among other things, the AssertionConsumerService URL, which is automatically opened for the user.

The existing user session in the browser is used for subsequent authentication. To access another Informatica web application configured to use SAML authentication, the user selects the LDAP security domain on the application log in page. It is not necessary for the user to supply a user name or password.

The user remains logged in to all Informatica web applications that are running in the same browser session. However, if the user logs out of a web application, the user is also logged out of other Informatica web applications running in the same browser session.

Before You Begin

Before you begin to configure Oracle Access Manager and Informatica, complete the tasks in this section.

Prerequisites

Perform these prerequisite tasks:

- Verify that you have deployed and configured Informatica version 10.5 in the same network domain as the Oracle implementation.
- Verify that you have a valid Oracle database and Oracle Access Manager license.
- Ask your IT administrator to assign IP addresses in the same network domain for the LDAP server.

Verify the Oracle Access Manager Environment

Verify that the Oracle administrator has installed and configured the following elements of an Oracle environment:

Oracle database

Install and create an Oracle database version 12.2.1 in the same network domain as the Active Directory server and the Informatica deployment. See the Oracle documentation:

- Downloading and installing Oracle
- Creating and configuring the Oracle database

After you create the database, change localhost to hostname and restart the OracleTNSListener service.

Oracle Access Manager

Install and configure the Oracle Access Manager identity provider in the same network domain as the Active Directory server and the Informatica deployment. See the Oracle documentation:

- Downloading and installing Oracle Access Manager

- Configuring Oracle Access Manager

Configure Oracle Access Manager

After you install and configure basic settings in Oracle Access Manager, perform the steps in this section to integrate Oracle Access manager with Informatica.

Step 1. Configure Active Directory as the Identity Store

The identity store is the repository of users and groups that Oracle Access Manager uses for authentication.

Configure Microsoft Active Directory (AD) as the identity store for Oracle Access Manager.

1. Create an Active Directory service account. See the Microsoft documentation.
2. Create a user identity store configuration in Oracle Access Manager. See the Oracle documentation.
3. Create an LDAP authentication module. See the Oracle documentation.
4. Create an authentication scheme. See the Oracle documentation.

Step 2. Install and configure Oracle HTTP Server and Web Gate

Oracle Access Manager requires a local web server and Oracle Web Gate to handle and forward authentication and authorization requests.

Read the Oracle documentation to download, install, and configure the Oracle HTTP web server and Oracle Web Gate:

- Download and install the HTTP Server and Oracle Web Gate
- Create and configure Oracle Web Gate

Step 3. Create a Service Provider Attribute Profile

Create a service provider attribute profile for the OAM connection to use.

The Service Provider Attribute Mapping Profile allows the administrator to map message attributes that are included in an incoming or outgoing message with OAM session attributes.

Use the Oracle Access Manager web interface to perform the following steps:

1. Browse to **Federation** > **Identity Provider Management Service Provider Attribute Profile** > **SP Attribute Profile**.
2. Populate the following properties:

Property Name	Description
Profile Name	Enter a name for the attribute profile.
Attribute Name	Name for the attribute in incoming and outgoing messages.
Always Send	Indicates if the attribute should be included in an outgoing assertion even when it has not been specifically requested. Verify that the Always Send option is selected.

3. In the **Attribute Mapping** panel, click **Create**.

4. Click **OK**, and then click **Save**.

Step 4. Create an Authentication Module

Create an authentication module to handle authentication requests. The authentication module matches the identity of the web application user with the user credentials in LDAP.

1. Browse to **Application Security** > **Plug-Ins** > **Authentication Module**.
2. Click **Search**.
3. Choose LDAP from the Authentication Module drop-down list, and then click **Edit**.
   The **LDAP** tab opens.
4. In the **LDAP Authentication Module** panel, click the **User Identity Store** drop-down list and select the identity store that you created.
5. Click **Apply**.

Step 5. Create an Authentication Scheme

Create an authentication scheme to define the authentication mechanism for users.

1. Browse to **Application Security** > **Access Manager** > **Authentication Scheme**.
2. Click **Search**.
3. Select **LDAPScheme** from the list of search results and click **Edit**.
4. Create a duplicate of LDAPScheme.
5. Populate the following properties for the authentication scheme:

Property	Description
Name	Name of the authentication scheme. Replace the name generated when you created the scheme as a duplicate with a unique name.
Authentication Level	The trust level of the authentication scheme used to protect transport of credentials from the user. Value is an integer between 0 (no trust) and 99 (highest level of trust). Verify the default value: 2.
Challenge Method	Set to FORM.
Authentication Module	Set to LDAP.
Challenge URL	URL associated with the challenge method. Verify the default value: `/pages/login.jsp`
Context type	Used to build the default URL for the embedded credential collector. Set to Default.
Context Value	Used to build the default URL for the embedded credential collector. Set to `/oam`.

   For more information about these properties, see the Oracle documentation.
6. Click **Apply**.
7. Click **Set as Default**.

Step 6. Create a Service Provider Connection

Define properties for the connection between Oracle and Informatica.

A service provider partner holds all the information that the Oracle Access Manager needs to communicate with the Informatica domain and the Administrator tool web application. Perform the following steps to define this connection:

1. Select the **Federation** tab and select **Identity Federation Management**.
2. Select **Create Service Provider Partner**.

Populate values in the following sections in the **Service Provider Partner** window.

General

1. Enter a name for the connection.
2. Verify that the **Enable Partner** option is checked.

Service Information

Populate the following properties:

Property	Description
Provider ID	Name to identify the service provider. For this integration, Informatica is the service provider. You can enter any name here. Make a note of the value that you enter for this property and use it when you "Update the Domain SAML Configuration" on page 9.
Assertion Consumer URL	Enter the URL for the Administrator Console. For example: `https://<host name>:<port number>`

Signing Certificate

Click **Choose File** to select a certificate, and then browse to and select the `OAM.cer` file. Click **OK**.

NameID Format

Populate the following properties:

Property	Description
NameID Format	Select **Unspecified** from the drop-down list.
NameID Value	Select **UserID Store Attribute** from the drop-down list, and then type `sAMAccountName` in the text box.

Mapping Options

Click **+ Add** to select the identity store profile you created in Step 1.

Advanced

For the **SSO Response Binding** property, select the HTTP POST option.

Step 7. Configure an Authentication Method and Certificate

Use a command line interface to configure an authentication method and a certificate for the Oracle WebLogic application server.

1. Open a command shell on the machine where the Oracle Access Manager instance runs.

2. Log in to the Oracle WebLogic server:`C:\Oracle\Middleware\Oracle_Home\oracle_common\common\bin>wlst.cmd`
3. Use the addSPPartnerAuthnMethod command to associate an authentication method and an Access Manager Authentication Scheme for Informatica.
   Specify the following three arguments:

Argument	Description
Partner Name	Name of the authentication scheme that you created in Step 5.
URN	Method that represents the authentication method. Enter the following identifier: `urn:oasis:names:tc:SAML:2.0:ac:classes:Password`
Connection name	Name of the connection that you defined in Step 6 under "General."

   For example:

```
/oam_domain/domainRuntime/> addSPPartnerAuthnMethod("AS", "urn:oasis:names:tc:SAML:
2.0:ac:classes:Password", "LDAPADScheme")
```

   See the Oracle documentation.

4. Use the updatePartnerProperty command to configure the certificate to use in authentication.
   Specify the following three arguments:

Argument	Description
partnerName	The ID of the partner to be updated.
partnerType	Partner type. Use "`sp`".
propName	Name of the property to configure.
propValue	Boolean. Set to TRUE. Default is false.
type	Data type of the property. Use "`boolean`".

   For example:

```
updatePartnerProperty(partnerName="AS", partnerType="sp",
propName="includecertinsignature",propValue="true",type="boolean")
```

   See the Oracle documentation.

Step 8. Add the Informatica Domain to the Hosts File

Edit the etc/hosts file on the OAM server to add the host name and IP address of the Informatica domain host.

Enable Oracle Access Manager for the Informatica Domain

Configure elements on the domain to enable integration with Oracle Access Manager.

Edit the hosts/etc File on Domain and Local Machines

Add the hostname and IP address of the Oracle Access Manager host to the `etc/hosts` file on the domain host and on client machines.

Verify that the Oracle Access Manager host information is added to every machine that runs a Developer tool that communicates with Oracle Access Manager.

Import the Truststore Certificate

The OAM.cer file validates the SAML assertion signature. Import the file to the Informatica domain truststore.

Get the OAM.cer file from the Oracle Access Manager and import it to the domain. Replace the default truststore file with this certificate file.

1. Enter the following URL and save the certificate file: `http://<Oracle Access Manager host>:<Oracle Access Manager port>/oamfed/idp/cert?id=osts_signing_sha256`
2. Use the keytool utility to import the certificate to the domain in the following location: `<Informatica home directory>/source/services/shared/security/infa_truststore.jks`
3. Restart the domain.

Update the Domain SAML Configuration

Use the infasetup updateDomainSAMLConfig command to enable SAML authentication on the domain and specify the identity provider URL.

For example: `./infasetup.sh updateDomainSAMLConfig -saml true -spid <provider ID value> -iu http://<Identity provider host>:<port>/oamfed/idp/samlv20`

Note: The example does not contain all required options. See the information about infasetup updateDomainSAMLConfig in the *Informatica Command Reference*.

The -spid option is required for Informatica releases 10.2.2 and higher. The option specifies the relying party trust name. Use the value that you provided in "Step 6. Create a Service Provider Connection" on page 7 under the Service Information section for the Provider ID property.

Optional: Configure the Certificate Alias on Gateway Nodes

If you use the assertion signing feature, update gateway nodes with the Oracle Access Manager certificate alias.

The certificate alias refers to the truststore certificate that enables signed assertion.

1. Run the following command to configure the certificate alias:
   `./infasetup.sh updateGatewayNode -saml true -asca <alias>`
2. After the command runs, recycle the Informatica domain.

Sync Oracle Access Manager LDAP Users

Sync users from the LDAP directory that resides on the Oracle Access Manager with the domain.

Run commands for the following tasks:

Add LDAP server connection properties

Run the addLDAPConnectivity command to add the connection information for an LDAP server. The LDAP server connection enables the Service Manager sets up the connection to an LDAP server and imports the user accounts of the LDAP security domains which are configured to this LDAP server.

Syntax: `infacmd.sh addLDAPConnectivity -dn <domain name> -un <domain user name> -pd <domain user password> -la <LDAP Server Address> -lt <LDAP types> -lp <LDAP principal> -lc <LDAP credential> -lcn <LDAP host configuration name>`

Add the identity provider namespace

Run the addNamespace command to add the identity provider user namespace to the domain. The command can filter the namespace by user name and group names.

Syntax: `infacmd.sh addNamespace -dn <domain name> -un <user name> -pd <password> -ns <namespace> -usb <usersearchbase> -uf <userfilter> -gsb <groupsearchbase> -gf <groupfilter> -lcn <LDAP host configuration name>`

Synchronize users

Run the SyncSecurityDomains command to synchronize security domain users and groups with LDAP users and groups.

Syntax: `infacmd.sh SyncSecurityDomains -dn <domain name> -un <user name> -pd <password> -sdn <security domain name> -sn <namespace to sync>`

See the *Informatica Command Reference* for more information about each command.

Complete Domain Configuration

Complete SAML configuration tasks in the Administrator tool.

Grant Privileges to the SAML User

Grant permissions to the domain to the identity provider SAML user:

1. In the Administrator tool, click the Security tab.
2. Search for the identity provider SAML user.
3. Assign Administrator permissions to the user.

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